

# Ming Li

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## Research Profile

Ph.D. student in Computer Science at the University of Maryland, advised by Prof. Tianyi Zhou. My research develops post-training, interpretability, and evaluation methods to understand and improve the behavior of large language models and agentic systems, with a focus on data-centric adaptation, reasoning behavior analysis, and agent evaluation. I have led 10+ first-author papers at ACL, NeurIPS, ICLR, EMNLP, and NAACL, served as an *ACL ARR Area Chair*, and received the **2026 Apple PhD Scholar in AI/ML Fellowship**.

## Education

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|---------------------------------------------------|------------------------------|----------------------|
| University of Maryland, Ph.D. in Computer Science | Advisor: Prof. Tianyi Zhou   | Aug 2023 – Present   |
| Texas A&M University, M.S. in Computer Science    | Advisor: Prof. Ruihong Huang | Aug 2021 – May 2023  |
| Xi'an Jiaotong University, B.S. Computer Science  |                              | Aug 2016 – June 2020 |

## Internship & Research Experience

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| MBZUAI, Visiting Researcher | Feb 2026 – May 2026 |
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- Studied agent evaluation and AI socialization in multi-agent environments.
- Developed evaluation protocols for coordination, social behavior, and emergent interaction among LLM agents.

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| Amazon, Applied Scientist Intern | May 2025 – Aug 2025 |
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- Developed reinforcement learning methods and evaluation pipelines for optimizing multi-turn LLM conversations.
- Studied reward design, trajectory-level feedback, and conversational behavior under iterative post-training.

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| Microsoft Research, Research Intern | Feb 2025 – May 2025 |
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- Studied interpretability and collaborative decoding mechanisms for LLM reasoning.
- Analyzed how decoding strategies affect reasoning quality, efficiency, and model behavior.

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| Adobe, Research Scientist/Engineer Intern | May 2024 – Nov 2024 |
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- Developed visual grounding and instruction-tuning methods for multimodal LLMs.
- Built data and evaluation pipelines for improving grounded instruction following in MLLMs.

## Research Areas

- **Data-centric AI and Post-training** [1–3]: Developed data-centric and algorithmic approaches for improving LLM adaptation, instruction following, and reasoning, including data selection, synthesis, reflection tuning, and data recycling.
- **Model Behavior and Interpretability** [4–6]: Analyzed how training data, post-training objectives, inference strategies, and interaction environments shape the behaviors of LLMs and agents, including reasoning, overthinking, and collaboration.
- **Agentic Systems and Evaluation Harnesses** [7–9]: Built methods, benchmarks, and environments for training and evaluating LLM agents, multi-agent societies, and agent-environment interaction.

## Selected Publications

- [1] Li, M., Zhang, Y., Li, Z., et al. From quantity to quality: Boosting LLM performance with self-guided data selection for instruction tuning. *NAACL'24*.
- [2] Li, M., Zhang, Y., He, S., et al. Superfiltering: Weak-to-strong data filtering for fast instruction-tuning. *ACL'24*.
- [3] Li, M., Chen, L., Chen, J., He, S., Gu, J., & Zhou, T. Selective reflection-tuning: Student-selected data recycling for LLM instruction-tuning. *ACL'24*.
- [4] Li, M., Fan, C., Cheng, Y., Feizi, S., & Zhou, T. Schoenfeld's anatomy of mathematical reasoning by language models. *ACL'26, Oral*.
- [5] Li, M., et al. What happened in LLMs layers when trained for fast vs. slow thinking: A gradient perspective. *ACL'25, Oral*.
- [6] Fan, C.\*, Li, M.\*, Sun, L., & Zhou, T. Missing premise exacerbates overthinking: Are reasoning models losing critical thinking skill? *COLM'25*.
- [7] Chen, Z.\*, Li, M.\*, Huang, Y., et al. ATLaS: Agent tuning via learning critical steps. *ACL'25*.
- [8] Li, M., Li, X., & Zhou, T. Does socialization emerge in AI agent society? A case study of Moltbook. *ACM CAIS'26*.
- [9] Li, X., Li, M., Xu, D., Chiang, W. L., Stoica, I., Hsieh, C. J., & Zhou, T. ClawEnvKit: Automatic Environment Generation for Claw-Like Agents.
- [10] Li, M., Jiao, H., Zhou, T., Zhang, N., Peters, S., & Lissitz, R. W. Item difficulty modeling using fine-tuned small and large language models. *Educational and Psychological Measurement, 2025*. *Best Research Award at e-Assessment Awards 2026*.

## Honors, Service, Talks, and Open Source

- **Honors:** **2026 Apple PhD Scholar in AI/ML Fellowship**; UMD Graduate School Dean's Fellowship
- **Academic Service:** ACL ARR Area Chair; reviewer for ACL, EMNLP, NAACL, ICLR, ICML, and NeurIPS
- **Talks, Tutorials, and Media:** ACM CIKM 2025 Tutorial on Data Synthesis for Data Mining; *Science News* interview on the emergence of AI societies
- **Open Source:** Cherry\_LLM, Reflection\_Tuning, Superfiltering, and other post-training projects, with 1000+ combined GitHub stars